Prime-Power Divisibility in Sun's A179524 Conjecture

An integral analytic digit operator and a Catalan diagonal reduction

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Abstract

For $a_n = \sum_k (-4)^k \binom{n}{k}^2 \binom{n-k}{k}^2$, put $S(n) = \sum_{k<n} (20k+17)a_k$. We prove $p^e \mid S(p^e m)$ for all $e, m \geq 1$ whenever $p = 5$ or $p \equiv 11, 13, 17, 19 \pmod{20}$, and retain the exact identity $v_2(S(n)) = v_2(n)$. An exact digit operator on restricted integral p-adic power series, with all carry boundaries included, permits simultaneous induction for arbitrary analytic weights and three distinguished moments. A separate one-moment cancellation treats $p = 5$. For every odd prime, the same operator propagates any finite chain of verified prime-power bases to all cofactors; consequently the full conjecture is equivalent to its prime-power instances. We also prove that the off-diagonal condition in the supplied negative-index manuscript follows from its diagonal condition, by an integral Catalan coordinate change. An exact telescoping certificate proves the previously guessed order-four recurrence. A fresh exact run through 50000, combined with propagation, gives an explicit proved domain of natural density greater than 0.9992. The remaining split-prime-power bases are not proved. No historical-priority claim is made for the general analytic lifting strategy or for the target-specific conclusions.

1 The unchanged target and the enlarged proved domain

Use zero for a binomial coefficient with an inadmissible lower index. Define

$$\begin{aligned} M(n,k) &= \binom{n}{k}\binom{n-k}{k}, \qquad a_n = \sum_{0 \leq 2k \leq n} (-4)^k M(n,k)^2, \\ S(n) &= \sum_{k=0}^{n-1} (20k+17)a_k. \end{aligned} \tag{1}$$

For $0 \leq 2k \leq n$, $M(n,k) = n!/[k!^2(n-2k)!]$. The initial values are $1, 1, -15, -143$. The original question is

$$n \mid S(n) \qquad (n \geq 1). \tag{2}$$

It is Sun's Conjecture 5.8(ii) in the 2010 source [1, printed p. 33], with $a_n = W_n(1, -4)$. OEIS A179524 attributes it to 1 July 2010 [2]. The numbered source is version 11; the final arXiv version was withdrawn to merge material into other papers.

The supplied manuscript [12] contributes an exact negative-index continuation and two-index formulations. The present note combines them with the previous CM prime-block argument, develops an exact analytic operator, proves propagation through all layers, and supplies a symbolic certificate for its order-four recurrence. All arguments needed for the stated results are included here.

Let

$$\mathcal{G} = \{5\} \cup \{p \text{ prime} : p \equiv 11, 13, 17, 19 \pmod{20}\}, \qquad N_0 = 50000.$$

For each odd prime p, put

$$e_p = \begin{cases} \infty, & p \in \mathcal{G}, \\ \max(\{1\} \cup \{e \geq 1 : p^e \leq N_0\}), & p \notin \mathcal{G}. \end{cases}$$

The minimum of a finite integer and ∞ is that integer.

Theorem 1.1 (Combined divisibility result)**.** *For every $n \geq 1$,*

$$v_2(S(n)) = v_2(n), \qquad 2^{v_2(n)} \prod_{\substack{p \mid n \\ p \text{ odd}}} p^{\min(v_p(n), e_p)} \mid S(n).$$

In particular, (2) holds for every n with $v_p(n) \leq e_p$ at each remaining odd prime. There is no exponent restriction at 2 or at any prime in $\mathcal{G}$.

The unrestricted prime families are proved analytically in Corollaries 6.5 and 7.2. The finite bottom chains are certified in Appendix B using the recurrence proved in Appendix A, and then propagated to every cofactor by Theorem 8.2. The exact 2-part and prime cases are proved below.

The remaining finite exponent bounds are particularly concrete:

remaining prime	all-cofactor exponent proved
3	9
7	5
23, 29	3
41, 43, 47, 61, 67, 83, 89, 101, 103, 107, 109	2
127, 149, 163, 167, 181, 223	2
$p > 223,\ p \equiv 1, 3, 7, 9 \pmod{20}$	1

For example, $3^9 \mid S(3^9 m)$ and $7^5 \mid S(7^5 m)$ hold for every positive m. These are not finite-cofactor tests.

Corollary 1.2 (A rigorous density bound)**.** *There is an explicit set of integers of natural density greater than 0.9992 on which (2) holds.*

Proof. Take the smaller set with arbitrary $v_2(n)$; at odd primes $p \leq 223$, allow arbitrary exponent if $p \in \mathcal{G}$, and exponent at most e_p otherwise; at every prime $p > 223$, allow exponent at most one. Define

$$\nu_p = \begin{cases} 1, & p \in \mathcal{G}, \\ 1 - p^{-(e_p+1)}, & p \notin \mathcal{G}. \end{cases}$$

The natural density of this set is

$$\delta_0 = \frac{8}{\pi^2} \prod_{\substack{3 \leq p \leq 223 \\ p \text{ prime}}} \frac{\nu_p}{1 - p^{-2}} = 0.99921205018052892427\ldots.$$

Existence follows by the Chinese remainder theorem for finitely many local conditions and the tail estimate $\sum_{p>y} p^{-2} \to 0$. The rigorous inequality $\delta_0 > 0.9992$ follows using $\pi < 355/113$ and exact rational arithmetic for the finite product; that rational comparison is in Appendix B. The larger domain in Theorem 1.1 also allows unrestricted exponents at inert primes above 223. This density measures a proved integer domain, not a proportion of the proof difficulty resolved. □

2 Prime divisibility and Lucas propagation

We first record two integer identities:

$$\binom{j}{m}^2 = \sum_{i=0}^{m} \binom{m}{i}\binom{m+i}{i}\binom{j}{m+i}, \tag{3}$$

$$\sum_{j<n}(20j+17)\binom{j}{t} = (20t+17)\binom{n}{t+1} + 20(t+1)\binom{n}{t+2}. \tag{4}$$

The first counts ordered pairs of m-subsets by their union size. The second follows from the hockey-stick identity and $j\binom{j}{t} = t\binom{j}{t} + (t+1)\binom{j}{t+1}$.

Lemma 2.1 (Ramanujan reduction)**.** *For an odd prime p, $h = (p-1)/2$, and $d = \lfloor p/4 \rfloor$,*

$$S(p) \equiv -(-1)^h \sum_{j=0}^{d} (20j+3)\frac{(4j)!}{j!^4(-1024)^j} \pmod{p}. \tag{5}$$

Proof. For $0 \le k \le h$, the polynomial $F_k(X) = \binom{X}{2k}^2$ is defined over $\mathbb{F}_p$ and satisfies $F_k(2k-1-X) = F_k(X)$. Thus

$$\sum_{r\in\mathbb{F}_p}(20r+17)F_k(r) = (20k+7)\sum_{r\in\mathbb{F}_p}F_k(r).$$

Apply (3) and sum over $0 \le r < p$. Modulo p, the only surviving binomial $\binom{p}{2k+i+1}$ has $2k+i = p-1$. Since $i = p-1-2k$ is even, its coefficient reduces to $\binom{2k}{p-1-2k}$. Therefore

$$S(p) \equiv \sum_{k=\lceil h/2\rceil}^{h} (20k+7)(-4)^k \binom{2k}{k}^2 \binom{2k}{2h-2k} \pmod{p}.$$

Put $k = h-j$. The needed congruences are

$$20k+7 \equiv -(20j+3), \quad (-4)^{h-j} \equiv (-1)^h(-4)^{-j},$$

$$\binom{p-1-2j}{h-j}^2 \equiv \binom{h+j}{2j}^2 \equiv \frac{\binom{2j}{j}^2}{256^j}, \qquad \binom{p-1-2j}{2j} \equiv \binom{4j}{2j}.$$

For the middle identity use $\binom{j-1/2}{2j} = (-1)^j\binom{2j}{j}/16^j$. All denominators are units modulo p. These formulas give (5). □

Sun's Theorem 1.2 [3] states, for $p > 3$,

$$\sum_{j=0}^{h}(20j+3)\frac{(4j)!}{j!^4(-1024)^j} \equiv (-1)^h p(2^{p-1}+2-(2^{p-1}-1)^2) \pmod{p^4}.$$

For $d < j \le h$, the numerator $(4j)!$ is divisible by p and $j!$ is not. Hence Lemma 2.1 gives $p \mid S(p)$. The remaining primes satisfy $S(2) = 54$ and $S(3) = -801$.

Lemma 2.2 (Digit congruence)**.** *For every odd prime p, $u \ge 0$, and $0 \le r < p$,*

$$a_{pu+r} \equiv a_u a_r \pmod{p}.$$

Proof. Write the inner index as $k = pv+s$, $0 \le s < p$. Multinomial Lucas makes $M(pu+r, pv+s)$ vanish modulo p unless $2s \le r$. In the latter case

$$M(pu+r, pv+s) \equiv M(u,v)M(r,s) \pmod{p}.$$

The high-digit range is exactly $0 \le 2v \le u$. Also $(-4)^{pv+s} \equiv (-4)^{v+s} \pmod{p}$. Summing the factored terms proves the claim. □

For every $m \geq 1$, it follows that

$$S(pm) \equiv S(p) \sum_{u<m} a_u \equiv 0 \pmod{p}. \tag{6}$$

The case $p = 2$ follows directly from $a_u \equiv 1 \pmod 2$. Consequently $\mathrm{rad}(n) \mid S(n)$ for every n. This argument alone does not increase the prime-power exponent.

3 The entire 2-part, with exact valuation

Theorem 3.1. *For every $n \geq 1$, $S(n)/n$ is an odd 2-adic integer. Thus $S(n) \neq 0$ and $v_2(S(n)) = v_2(n)$.*

Proof. Rewrite $M(j,k) = \binom{j}{2k}\binom{2k}{k}$. For a fixed $k \geq 1$, use (3) with $m = 2k$, followed by (4), and divide by n. Its contribution is an integer linear combination of

$$(-4)^k \binom{2k}{k}^2 \left(\frac{20t+17}{t+1}\binom{n-1}{t} + \frac{20(t+1)}{t+2}\binom{n-1}{t+1} \right), \qquad 2k \leq t \leq 4k.$$

The prefactor has 2-adic valuation at least $2k+2$, because $\binom{2k}{k} = 2\binom{2k-1}{k-1}$. Both displayed denominators are at most $4k+2 < 2^{2k+2}$. Every such term therefore has valuation at least one. All sums are finite. The $k = 0$ contribution is

$$\frac{1}{n}\sum_{j<n}(20j+17) = 10n+7,$$

which is odd. This proves the assertion, including $n = 1$. □

4 A prime-square transfer valid for every cofactor

Put $f(n,k) = (-4)^k M(n,k)^2$, and define

$$L_n = \sum_k k f(n,k), \qquad U(N) = \sum_{n<N} a_n, \quad T(N) = \sum_{n<N} n a_n, \quad V(N) = \sum_{n<N} L_n.$$

Thus $S(N) = 20T(N) + 17U(N)$. For an odd prime p, let

$$Q_p = \frac{(-4)^{p-1}-1}{p},$$
$$A_r = \sum_s f(r,s)(H_r - H_{r-2s}), \qquad B_r = \sum_s f(r,s)(H_{r-2s} - H_s) \quad (0 \leq r < p),$$

where $H_0 = 0$, $H_j = \sum_{i=1}^{j} 1/i$, and only admissible s are included. These harmonic numbers are p-integral.

Lemma 4.1 (First-order digit expansion)**.** *For every $q \geq 0$ and $0 \leq r < p$,*

$$a_{pq+r} \equiv a_q a_r + p\{2qa_qA_r + L_q(Q_p a_r + 4B_r)\} \pmod{p^2}. \tag{7}$$

Proof. Let $E_p(N) = \prod_{1\leq i\leq N,\ p\nmid i} i$. The exact factorization $N! = p^{\lfloor N/p\rfloor}\lfloor N/p\rfloor! E_p(N)$, together with $\sum_{i=1}^{p-1} i^{-1} = 0$ in $\mathbb{F}_p$, gives

$$E_p(pq+r) \equiv ((p-1)!)^q r!(1+pqH_r) \pmod{p^2}.$$

Write $k = pj + s$. If $2s > r$, Lucas gives $p \mid M(pq + r, k)$, so its square vanishes modulo p^2. Otherwise the factorial factorization gives

$$M(pq + r, pj + s) \equiv M(q, j)M(r, s)[1 + p\{qH_r - 2jH_s - (q - 2j)H_{r-2s}\}] \pmod{p^2}.$$

Only E_p-factors and small factorials are inverted. The integer $M(q, j)$ is retained exactly, so higher carries cause no problem. Finally

$$(-4)^{pj+s} \equiv (-4)^{j+s}(1 + pjQ_p) \pmod{p^2}.$$

Square the multinomial congruence, multiply, and sum over j, s. □

Theorem 4.2 (Prime-square transfer)**.** *For every odd prime p and every positive integer m,*

$$S(p^2m) \equiv U(m)S(p^2) \pmod{p^2}. \tag{8}$$

Proof. Write $w_r = 20r + 17$, $U_p = U(p)$, $T_p = T(p)$, and $V_p = V(p)$. Summing (7) with weight $20(pq + r) + 17$ gives

$$S(pM) \equiv S(p)U(M) + pC_pT(M) + pD_pV(M) \pmod{p^2}, \tag{9}$$

where

$$C_p = 20U_p + 2\sum_{r<p} w_rA_r, \qquad D_p = 4\sum_{r<p} w_rB_r.$$

The omitted term is exactly $pQ_pS(p)V(M)$; it vanishes because $p \mid S(p)$, proved in Section 2. Lucas also gives $L_{pu+r} \equiv a_uL_r \pmod{p}$, since the index $pv + s$ reduces to s. Hence

$$U(pm) \equiv U_pU(m), \quad T(pm) \equiv T_pU(m), \quad V(pm) \equiv V_pU(m) \pmod{p}.$$

Substitute $M = pm$ into (9). Its three coefficients $S(p), pC_p, pD_p$ are divisible by p, so these substitutions are valid modulo p^2. The result is

$$S(p^2m) \equiv U(m)[S(p)U_p + pC_pT_p + pD_pV_p].$$

The bracket is $S(p^2)$ modulo p^2 by (9) with $M = p$. □

Remark 4.3 (What a finite base verifies). For a fixed prime p, an exact verification of $p^2 \mid S(p^2)$ now proves $p^2 \mid S(p^2m)$ for *every* m. It is a proof for an unbounded cofactor, but not a verification over all primes. The theorem also gives no p^3 conclusion.

5 An infinite prime-square family from complex multiplication

Theorem 5.1. *If $p \equiv 11, 13, 17, 19 \pmod{20}$ is prime, then*

$$p^2 \mid S(p^2m) \qquad (m \geq 1).$$

Proof. For an indeterminate c, define

$$W_r(c) = \sum_k c^k M(r, k)^2, \qquad \mathcal{U}_p(c) = \sum_{r<p} W_r(c).$$

Set $h = (p - 1)/2$ and $d = \lfloor p/4 \rfloor$. The binomial-square summation in Section 2, followed by $k = h - j$, gives the polynomial identity

$$\mathcal{U}_p(c) \equiv c^h \sum_{j=0}^{d} \frac{(4j)!}{j!^4(256c)^j} \pmod{p}. \tag{10}$$

All apparent negative powers cancel. The finite Legendre identity

$$P_d(t)^2 = \sum_{j=0}^{d} \binom{d}{j}\binom{d+j}{j}\binom{2j}{j}\left(\frac{t^2-1}{4}\right)^j$$

and $\{-d, d+1\} = \{1/4, 3/4\}$ in $\mathbb{F}_p$ imply

$$\mathcal{U}_p(c) \equiv c^h P_d(t)^2, \qquad t^2 = 1 - \frac{1}{c}. \tag{11}$$

This is an identity in the parameter, not merely an equality at a particular numerical value.

Take $t_0 = \sqrt{5}/2$, corresponding to $c = -4$, and consider

$$E_{t_0}: \quad y^2 = x\left(x^2 + 2x + \frac{1-t_0}{2}\right).$$

Its invariants are

$$\Delta = 8(t_0-1)^2(t_0+1), \qquad j(E_{t_0}) = \frac{64(3t_0+5)^3}{(t_0-1)^2(t_0+1)} = 632000 + 282880\sqrt{5}.$$

Duke [4] identifies this value as $j(i\sqrt{5})$; hence the curve has complex multiplication of discriminant -20. The other sign of t_0 gives the conjugate CM value. The coefficients are integral away from 2, and $(t_0-1)(t_0+1) = 1/4$, so the model has good reduction at every prime above an odd rational prime.

The stated residue classes are exactly $\left(\frac{-5}{p}\right) = -1$. Deuring's reduction theorem therefore makes this reduction supersingular; see [5]. This holds over $\overline{\mathbb{F}}_p$, including when $t_0 \notin \mathbb{F}_p$.

For completeness, the Hasse coefficient can be read off directly. Let $b = (1-t)/2$. The coefficient of x^{p-1} in $[x(x^2+2x+b)]^h$ is

$$\sum_{k=0}^{d} \frac{h!}{k!^2(h-2k)!}\, 2^{h-2k} b^k = 2^h \sum_{k=0}^{d} \frac{(-h/2)_k((1-h)/2)_k}{k!^2} b^k \equiv 2^h P_d(t) \pmod{p}.$$

Here $(z)_k$ is the rising factorial. The standard Hasse-invariant criterion [6, Theorem V.4.1(a)] therefore gives $P_d(t_0) = 0$. Equation (11) yields

$$U_p = \mathcal{U}_p(-4) \equiv 0 \pmod{p}. \tag{12}$$

There are two elementary ways to obtain the other moments. The reflection used in Section 2 also gives, for each k,

$$\sum_{r<p} r\binom{r}{2k}^2 \equiv \left(k - \frac{1}{2}\right)\sum_{r<p}\binom{r}{2k}^2 \pmod{p}.$$

Thus $T_p \equiv V_p - U_p/2 \pmod{p}$. Since $20T_p + 17U_p = S(p) \equiv 0 \pmod{p}$ and $p \neq 2, 5$, (12) implies $T_p = V_p = 0$ in $\mathbb{F}_p$. Alternatively, differentiate (11): the square factor gives $\mathcal{U}_p'(-4) = 0$ because $t_0 \neq 0$; then $V_p = -4\mathcal{U}_p'(-4) = 0$.

Finally the bracket in the proof of Theorem 4.2 vanishes modulo p^2: $p \mid S(p)$ and U_p, T_p, V_p all vanish modulo p. This proves the theorem for every cofactor m. □

6 An integral analytic digit operator

The fixed-modulus expansion in Section 4 is replaced here by an exact identity in a complete ring. This keeps the entire remainder inside a controlled integral space and permits induction through every prime-power exponent.

Fix a prime $p \geq 5$ and $c \in \mathbb{Z}_p^\times$. Write

$$f_c(n,k) = c^k \left(\frac{n!}{k!^2 (n-2k)!} \right)^2 \quad (0 \leq 2k \leq n), \qquad f_c(n,k) = 0 \quad \text{otherwise.}$$

Let $\mathcal{A}_p = \mathbb{Z}_p\langle X, Y \rangle$ denote the restricted power-series ring: coefficients are p-adic integers and tend to zero as total degree tends to infinity. It is complete for the Gauss norm, and its elements may be evaluated at any pair in $\mathbb{Z}_p^2$. Define

$$I_N(F) = \sum_{n=0}^{N-1} \sum_{0 \leq 2k \leq n} f_c(n,k) F(n,k), \quad U_c(N) = I_N(1), \quad T_c(N) = I_N(X), \quad V_c(N) = I_N(Y).$$

For $c = -4$, these agree with the earlier U, T, V.

Lemma 6.1 (Integral interpolation of the unit factorial)**.** *Put*

$$E_p(N) = \prod_{\substack{1 \leq i \leq N \\ p \nmid i}} i, \qquad W = (p-1)!.$$

There are units $\mathcal{H}_r(X) \in 1 + p\mathbb{Z}_p\langle X \rangle$, $0 \leq r < p$, such that, for every integer $q \geq 0$,

$$E_p(pq + r) = W^q r!\, \mathcal{H}_r(q), \qquad \mathcal{H}_r(X) \equiv 1 + pH_r X \pmod{p^2}.$$

Here $H_r = \sum_{i=1}^{r} 1/i$, with $H_0 = 0$. The first identity is exact.

Proof. Let $\eta_j = \sum_{i=1}^{p-1} i^{-j}$, and set

$$B(X) = \prod_{i=1}^{p-1} (1 + pX/i), \qquad \log B(X) = \sum_{j \geq 1} \ell_j X^j, \quad \ell_j = \frac{(-1)^{j-1} p^j \eta_j}{j}.$$

Let $S_j(q) = \sum_{h=0}^{q-1} h^j$. Its polynomial continuation is

$$S_j(X) = \sum_{a=0}^{j} \left\{ {j \atop a} \right\} \frac{(X)_{\underline{a+1}}}{a+1},$$

where braces denote Stirling numbers of the second kind and the underlined subscript denotes a falling factorial. Consequently

$$S_j(X) \in p^{-\lfloor \log_p(j+1) \rfloor} \mathbb{Z}_p[X].$$

Define $A(X) = \sum_{j \geq 1} \ell_j S_j(X)$. We claim $A \in p^3 \mathbb{Z}_p\langle X \rangle$. Indeed, $\eta_2 \equiv 0 \pmod{p}$, and pairing i with $p - i$ gives $2\eta_1 \equiv -p\eta_2 \pmod{p^2}$. Thus the terms $j = 1, 2$ have coefficient valuation at least three. For $j \geq 3$ every coefficient of the j-th term has valuation at least

$$j - v_p(j) - \lfloor \log_p(j+1) \rfloor \geq 3,$$

and this bound tends to infinity. For $j = 3, 4, 5$ the inequality is immediate, including the worst cases $p = 5, j = 4, 5$. For $j \geq 6$, use $v_p(j) \leq \lfloor \log_5(j+1) \rfloor$ and $2\lfloor \log_5(j+1) \rfloor \leq j - 3$. The latter is direct below 24; if $t = \lfloor \log_5(j+1) \rfloor \geq 2$, then $j \geq 5^t - 1$ and $2t \leq 5^t - 4$.

It follows that $\mathcal{E}(X) = \exp A(X)$ belongs to $1 + p^3\mathbb{Z}_p\langle X\rangle$: the k-th exponential term has Gauss valuation at least $3k - v_p(k!)$, tending to infinity. Set

$$\mathcal{H}_r(X) = \mathcal{E}(X)\prod_{i=1}^{r}(1 + pX/i).$$

The first-order formula follows. At a nonnegative integer q, convergence permits evaluation and interchange with the finite sum:

$$A(q) = \sum_{h=0}^{q-1}\log B(h), \qquad \mathcal{E}(q) = \prod_{h=0}^{q-1} B(h).$$

Splitting the unit factorial into its q complete blocks and its last r factors proves the exact identity. We interpolate only the normalized factor; no p-adic interpolation of W^q is required. □

Lemma 6.2 (Exact carry-aware kernel)**.** *For $0 \le r, s < p$, put*

$$d = \left\lceil \frac{2s - r}{p} \right\rceil \in \{0, 1, 2\}, \qquad t = r - 2s + dp, \qquad z = X - 2Y.$$

Then $0 \le t < p$. Define

$$\mathcal{C}(Y) = \exp(Y\log(c^{p-1})), \qquad Q_c = \frac{c^{p-1} - 1}{p},$$

and

$$K_{rs}(X, Y) = p^{2d}(z)_{\underline{d}}^2 \frac{c^s W^{2d} r!^2}{s!^4 t!^2} \frac{\mathcal{C}(Y)\mathcal{H}_r(X)^2}{\mathcal{H}_s(Y)^4\mathcal{H}_t(X - 2Y - d)^2}.$$

This is an element of $\mathcal{A}_p$. For every $q \ge 0$ and $0 \le 2j \le q$,

$$f_c(pq + r, pj + s) = f_c(q, j)K_{rs}(q, j).$$

The equality includes inadmissible original summands, interpreted as zero. Furthermore,

$$K_{rs}(0, 0) = f_c(r, s).$$

If $d \ge 1$, then $K_{rs} \in p^2\mathcal{A}_p$; if $d = 0$, then

$$K_{rs}(X, Y) \equiv f_c(r, s)\big[1 + p\{2X(H_r - H_t) + Y(Q_c + 4H_t - 4H_s)\}\big] \pmod{p^2}.$$

Proof. Since $c^{p-1} \in 1 + p\mathbb{Z}_p$, the exponential defining $\mathcal{C}$ converges and has integral restricted coefficients: $k - v_p(k!) \to \infty$. It is congruent to $1 + pQ_cY$ modulo p^2. All factorial denominators in the kernel are units. Affine substitutions with integral coefficients preserve restricted series, and the $\mathcal{H}$'s are units.

Write $q_0 = q - 2j$. If $q_0 \ge d$, use $N! = p^{\lfloor N/p\rfloor}\lfloor N/p\rfloor!E_p(N)$ and Lemma 6.1. Before squaring, the factorial ratio equals

$$p^d(q_0)_{\underline{d}} \frac{q!}{j!^2 q_0!} \frac{W^d r!}{s!^2 t!} \frac{\mathcal{H}_r(q)}{\mathcal{H}_s(j)^2\mathcal{H}_t(q_0 - d)}.$$

The powers W^q cancel to the fixed factor W^d. Also $c^{pj+s} = c^j c^s\mathcal{C}(j)$, proving the equality. If $0 \le q_0 < d$, the original term is zero and $(q_0)_{\underline{d}} = 0$, so the same equality holds. The denominator at the negative argument $q_0 - d$ remains an analytic unit. No j with $2j > q$ can contribute, since $p(q - 2j) + r - 2s < 0$. This proves that a common high-index range is valid for every pair of low digits. The remaining assertions follow from the kernel and the first-order unit expansions. □

Proposition 6.3 (Exact analytic block identity)**.** *For $F \in \mathcal{A}_p$, define*

$$(\mathcal{T}_p F)(X,Y) = \sum_{r,s=0}^{p-1} K_{rs}(X,Y) F(pX+r, pY+s).$$

Then $\mathcal{T}_p F \in \mathcal{A}_p$, and

$$I_{pN}(F) = I_N(\mathcal{T}_p F) \qquad (N \geq 1).$$

There exist $\alpha_F, \beta_F \in \mathbb{Z}_p$ and $R_F \in \mathcal{A}_p$ such that

$$\mathcal{T}_p F = I_p(F) + p\alpha_F X + p\beta_F Y + p^2 R_F. \tag{13}$$

In particular, the following is an exact equality, not a truncated congruence:

$$I_{pN}(F) = I_p(F) U_c(N) + p\alpha_F T_c(N) + p\beta_F V_c(N) + p^2 I_N(R_F). \tag{14}$$

Proof. Partition the two summation indices into their high and low digits and apply Lemma 6.2. For the expansion use

$$F(pX+r, pY+s) \equiv F(r,s) + pXF_X(r,s) + pYF_Y(r,s) \pmod{p^2}.$$

This holds first for polynomials and then for restricted series by Gauss-norm continuity. The kernel has an affine first-order part. Its exact value at $(0,0)$ makes the constant term exactly $I_p(F)$. The residual after removing the constant and the affine first-order part is divisible by p^2 in the restricted ring. Evaluating the resulting identity gives (14). □

Theorem 6.4 (Simultaneous valuation induction)**.** *Suppose $U_c(p), T_c(p), V_c(p) \in p\mathbb{Z}_p$. For every $e \geq 1$, $m \geq 1$, and $F \in \mathcal{A}_p$,*

$$I_{p^e m}(F) \in p^{e-1}\mathbb{Z}_p.$$

If in addition $I_p(F) \in p\mathbb{Z}_p$, then

$$I_{p^e m}(F) \in p^e \mathbb{Z}_p.$$

In particular this stronger conclusion applies to $F = 1, X, Y$.

Proof. Fix m. At $e = 1$, the first assertion is integrality. If $I_p(F) \in p\mathbb{Z}_p$, every term of (14) with $N = m$ is divisible by p, proving the second assertion.

Assume both assertions at $e-1$, and set $N = p^{e-1}m$. By the hypothesis and the stronger induction assertion, $U_c(N), T_c(N), V_c(N) \in p^{e-1}\mathbb{Z}_p$. By the universal assertion, $I_N(R_F) \in p^{e-2}\mathbb{Z}_p$. In (14), the first term is therefore divisible by p^{e-1} and the other three by p^e. If $I_p(F)$ is divisible by p, the first term is also divisible by p^e. This proves both assertions simultaneously, for every restricted test function and every exponent. □

Corollary 6.5 (All inert prime powers)**.** *For every prime $p \equiv 11, 13, 17, 19 \pmod{20}$, every $e \geq 1$, and every $m \geq 1$,*

$$p^e \mid U(p^e m), \qquad p^e \mid T(p^e m), \qquad p^e \mid V(p^e m), \qquad \boxed{p^e \mid S(p^e m)}.$$

Proof. Take $c = -4$. The CM and Legendre-square argument in Theorem 5.1 proves $U(p) \equiv 0 \pmod{p}$. Differentiating that coefficientwise square identity gives $V(p) \equiv 0 \pmod{p}$; the reflection relation gives $T(p) \equiv V(p) - U(p)/2 \equiv 0 \pmod{p}$. Apply Theorem 6.4, and use $S = 20T + 17U$. □

7 The entire 5-part

The prime 5 is handled by a different cancellation in the same exact operator; its first three prime-block moments do not all vanish.

Theorem 7.1 (One-moment criterion)**.** *Suppose $p \geq 5$ and*

$$\mathcal{T}_p(1) = C + p^2 R, \qquad C \in p\mathbb{Z}_p, \quad R \in \mathcal{A}_p.$$

Then for all $e, m \geq 1$,

$$I_{p^e m}(F) \in p^{e-1}\mathbb{Z}_p \quad (F \in \mathcal{A}_p), \qquad U_c(p^e m) \in p^e \mathbb{Z}_p.$$

If F is constant modulo p, then $I_{p^e m}(F) \in p^e \mathbb{Z}_p$.

Proof. The assertions at $e = 1$ follow from integrality and $U_c(pm) = CU_c(m) + p^2 I_m(R)$. Assume the assertions at $e - 1$. In (14), the constant term is divisible by p^{e-1}, the two affine terms by $p \cdot p^{e-2}$, and the residual by $p^2 \cdot p^{e-2}$. This proves the universal bound at e. For $F = 1$, use instead $U_c(pN) = CU_c(N) + p^2 I_N(R)$; both terms are divisible by p^e. Finally, if $F = a + pG$, then $I(F) = aU_c + pI(G)$, giving the last claim. □

Corollary 7.2. *For every $e \geq 1$ and $m \geq 1$,* $\boxed{5^e \mid S(5^e m)}$.

Proof. Set $p = 5, c = -4$. The harmonic residues are $(H_0, H_1, H_2, H_3, H_4) = (0, 1, 4, 1, 0)$ modulo 5, and $Q_c = 51 \equiv 1$. For each no-carry pair put $t = r - 2s$, $f = f_{-4}(r, s)$. The kernel shows that the two first-order coefficients of $\mathcal{T}_5(1)$ are the sums of $2f(H_r - H_t)$ and $f[Q_c + 4(H_t - H_s)]$. Their complete row-by-row evaluation is

r	$\sum_s f$	$\sum_s 2f(H_r - H_t)$	$\sum_s f[Q_c + 4(H_t - H_s)]$
0	1	0	1
1	1	0	0
2	0	2	0
3	2	0	1
4	1	3	3
total modulo 5	0	0	0

All carry terms are divisible by 25. The exact constant is $U(5) = 1 + 1 - 15 - 143 + 1 = -155$. Hence

$$\mathcal{T}_5(1) = -155 + 25R, \qquad R \in \mathcal{A}_5.$$

Apply Theorem 7.1 to $F = 20X + 17$, which is constant modulo 5. Notice that $T(5) \equiv V(5) \equiv 1 \pmod 5$; no vanishing of those moments is being assumed. □

8 Propagation through every finite chain of prime-power bases

The exact operator also removes a separate propagation conjecture from the reduction to prime powers. Only its constant reduction modulo p is needed here, so the argument includes $p = 3$.

Lemma 8.1. *For every odd prime p and $c \in \mathbb{Z}_p^\times$, the exact kernel and block identity of Lemma 6.2 and Proposition 6.3 exist over $\mathcal{A}_p$, with*

$$\mathcal{T}_p F = I_p(F) + pR_F, \qquad R_F \in \mathcal{A}_p, \quad R_F(0, 0) = 0.$$

At $p = 3$ no affine assertion modulo p^2 is made.

Proof. For $p \geq 5$ this is already proved. For $p = 3$, the construction of $A(X)$ in Lemma 6.1 still converges, now in $3\mathbb{Z}_3\langle X\rangle$. The term $j = 1$ has valuation at least two, since $\eta_1 = 3/2$. For $j \geq 2$ the Stirling bound gives

$$j - v_3(j) - \lfloor \log_3(j+1) \rfloor \geq 1,$$

with a bound tending to infinity. Check $j = 2, 3$ directly; for $4 \leq j \leq 7$ the bound is at least three, and for $j \geq 8$ use $2\lfloor \log_3(j+1) \rfloor \leq j - 1$. The exponential of an element of $3\mathbb{Z}_3\langle X\rangle$ lies in $1 + 3\mathbb{Z}_3\langle X\rangle$, because $k - v_3(k!) \to \infty$. Thus all exact interpolation and carry identities remain valid, and $K_{rs} \equiv f_c(r, s) \pmod p$. Finally $F(pX + r, pY + s) \equiv F(r, s) \pmod p$, proving the assertion. □

Theorem 8.2 (Propagation from preceding bottom layers)**.** *Let p be odd, $c \in \mathbb{Z}_p^\times$, $F \in \mathcal{A}_p$, and $e \geq 1$. Suppose*

$$I_{p^j}(F) \in p^j\mathbb{Z}_p \qquad (1 \leq j < e).$$

Then, for every $m \geq 1$,

$$I_{p^e m}(F) \equiv I_{p^e}(F)U_c(m) \pmod{p^e}.$$

If the base divisibility also holds at $j = e$, then $I_{p^e m}(F) \in p^e\mathbb{Z}_p$ for all m.

Proof. Since $I_1(G) = G(0, 0)$, repeated exact block decomposition gives

$$(\mathcal{T}_p^j F)(0, 0) = I_{p^j}(F).$$

Inductively, if the bottom divisibilities hold through $j - 1$, write $\mathcal{T}_p^{j-1}F = p^{j-1}G$. Lemma 8.1 gives

$$\mathcal{T}_p^j F = p^{j-1}I_p(G) + p^j R_G = I_{p^j}(F) + p^j R_G.$$

If the next bottom value is divisible by p^j, the entire iterate is in $p^j\mathcal{A}_p$. Applying this induction through $e - 1$ and then once more yields $\mathcal{T}_p^e F = I_{p^e}(F) + p^e R_G$. Apply I_m to obtain the asserted congruence. For $e = 1$ the hypothesis is empty and the same argument applies. □

Corollary 8.3 (Reduction of the original conjecture)**.** *Sun's original assertion $n \mid S(n)$ for every $n \geq 1$ is equivalent to its restriction $p^e \mid S(p^e)$ to all prime powers.*

Proof. Necessity is immediate. For sufficiency, use $c = -4$ and $F = 20X + 17$ in Theorem 8.2 at each odd prime divisor $p^e \| n$. The 2-part is Theorem 3.1. The resulting prime-power divisors are pairwise coprime. □

The supplied manuscript [12] formulates an unconditional all-layer congruence as a separate conjecture. Theorem 8.2 proves precisely the propagation needed once preceding bottom layers are known, so that separate conjecture is no longer needed for the prime-power reduction. We do not claim its unconditional version without those preceding hypotheses. The still missing assertions are the unrestricted bottom values at the remaining split primes.

9 Catalan coordinates remove the off-diagonal condition

The negative-index continuation in the supplied manuscript [12] introduces a two-variable divisibility condition. Its two stated one-variable conditions are not independent: the diagonal condition alone implies the off-diagonal one. We prove this by an integral change of variables. The diagonal condition itself remains unproved in full.

Put

$$\gamma_l = (-4)^l \binom{2l}{l}^2, \qquad G(x) = \sum_{l \geq 0} \gamma_l x^{2l}, \qquad \rho = \frac{uv}{1 + u + v}, \qquad \Lambda = \frac{(1+u)(1+v)}{uv},$$

and define

$$A = \mathbb{Z}[[u, v]], \qquad \theta_u = u\frac{\partial}{\partial u}, \quad \theta_v = v\frac{\partial}{\partial v}, \qquad \mathcal{D}_A = \theta_u A + \theta_v A, \qquad \Psi = (3\rho + 20\rho^2)G(uv).$$

Constant terms are taken in the ring $\bigcup_{N\geq 0}(uv)^{-N}A$, so every operation below is coefficientwise well-defined.

Lemma 9.1 (The negative-index reduction)**.** *For every positive integer N,*

$$S(N) = -\operatorname{CT}(\Lambda^N \Psi) + 20N \operatorname{CT}(\Lambda^N \rho G(uv)). \tag{15}$$

Consequently, $\Psi \in \mathcal{D}_A$ implies $N \mid S(N)$ for every $N \geq 1$.

Proof. First,

$$\operatorname{CT}(\Lambda^k G(uv)) = \sum_l \gamma_l \binom{k}{2l}^2 = a_k.$$

The polynomial identity

$$(T-1)^2 \sum_{k=0}^{N-1} (20k + 17)T^k = (3T + 17)(1 - T^N) + 20NT^N(T - 1)$$

may be evaluated at $T = \Lambda$, since $(\Lambda - 1)^{-1} = \rho$. Multiplication by $G(uv)$ and taking constant terms give (15); the constant term of Ψ is zero. Equivalently, the negative-index generating series is

$$\sum_{n\geq 1} (20n - 17)\Lambda^{-n} = 3\rho + 20\rho^2.$$

If $\Psi = \theta_u J_1 + \theta_v J_2$ with integral series J_i, then formal integration by parts gives

$$\operatorname{CT}(\Lambda^N \theta_u J_1) = -N \operatorname{CT}(\Lambda^{N-1}\theta_u(\Lambda) J_1) \in N\mathbb{Z},$$

and the same holds for the v-derivative. This proves the implication. □

For a series $F \in A$, write

$$\operatorname{Diag} F(x) = \sum_{n\geq 0} [u^n v^n] F \, x^n.$$

We use the elementary coefficient criterion

$$F \in \mathcal{D}_A \iff [u^0 v^0]F = 0 \text{ and } \gcd(r, s) \mid [u^r v^s]F \quad ((r, s) \neq (0, 0)). \tag{16}$$

Indeed, each coefficient of a divergence has the form $rj_1 + sj_2$. Conversely, Bezout's identity permits such integers j_1, j_2 to be chosen separately for every coefficient.

Theorem 9.2 (Integral diagonal criterion)**.** *Let $R(x) \in \mathbb{Z}[[x]]$ and $\alpha, \beta \in \mathbb{Z}$. For*

$$F(u, v) = (\alpha\rho + \beta\rho^2)R(uv),$$

the following conditions are equivalent:

1. *$F \in \mathcal{D}_A$;*
2. *$n \mid [u^n v^n]F$ for every $n \geq 1$.*

The statement holds over $\mathbb{Z}$, including at the prime 2.

Proof. Only the reverse implication requires proof. Let $C(x) \in 1 + x\mathbb{Z}[[x]]$ be the Catalan series, characterized by $C = 1 + xC^2$. Define

$$U = uC(uv), \qquad V = vC(uv), \qquad y = UV.$$

This is an integral formal automorphism with inverse

$$u = \frac{U}{1+y}, \qquad v = \frac{V}{1+y}, \qquad x = uv = \frac{y}{(1+y)^2}, \qquad C(x) = 1 + y. \tag{17}$$

In these coordinates,

$$1 + u + v = \frac{(1+U)(1+V)}{1+y}, \qquad \rho = \frac{y}{(1+y)(1+U)(1+V)}.$$

The logarithmic differential matrix of the inverse is

$$\begin{pmatrix} d\log u \\ d\log v \end{pmatrix} = \frac{1}{1+y}\begin{pmatrix} 1 & -y \\ -y & 1 \end{pmatrix}\begin{pmatrix} d\log U \\ d\log V \end{pmatrix}.$$

Its determinant $J(y) = (1-y)/(1+y)$ is an integral power-series unit; the matrix and its inverse have integral coefficients. Exact logarithmic two-forms are therefore preserved in both directions. Explicitly,

$$F\, d\log u \wedge d\log v = d\big(J_1\, d\log v - J_2\, d\log u\big)$$

is equivalent to $F = \theta_u J_1 + \theta_v J_2$, and pulling back the one-form keeps both coefficients integral. Hence $F \in \mathcal{D}_A$ if and only if

$$\widetilde{F}(U,V) = J(y)F\left(\frac{U}{1+y}, \frac{V}{1+y}\right)$$

is an integral formal divergence in U, V.

There are integral series

$$A_1(y) = \frac{\alpha y(1-y)}{(1+y)^2} R\left(\frac{y}{(1+y)^2}\right), \qquad A_2(y) = \frac{\beta y^2(1-y)}{(1+y)^3} R\left(\frac{y}{(1+y)^2}\right)$$

such that

$$\widetilde{F}(U,V) = \frac{A_1(y)}{(1+U)(1+V)} + \frac{A_2(y)}{(1+U)^2(1+V)^2}. \tag{18}$$

Write $A_1(y) = \sum_k A_{1,k} y^k$, $A_2(y) = \sum_k A_{2,k} y^k$. For $r, s \geq 1$, put $q = \min(r,s)$ and $d = \gcd(r,s)$. Expanding the separated denominators gives

$$[U^r V^s]\widetilde{F} = (-1)^{r+s} \sum_{k=0}^{q} \big(A_{1,k} + A_{2,k}(r-k+1)(s-k+1)\big).$$

Since r, s, q are all divisible by d,

$$[U^r V^s]\widetilde{F} \equiv (-1)^{r+s}[U^q V^q]\widetilde{F} \pmod{d}. \tag{19}$$

Thus divisibility of the diagonal coefficients is sufficient for all the required off-diagonal divisibilities in these coordinates.

It remains to justify diagonal divisibility after the change of variables. The radial substitution (17) preserves the difference of the two exponents. Therefore only old diagonal monomials contribute to the new diagonal, and

$$\operatorname{Diag}\widetilde{F}(y) = J(y)\ \operatorname{Diag} F\left(\frac{y}{(1+y)^2}\right).$$

By the assumed diagonal divisibility there is $H(x) \in x\mathbb{Z}[[x]]$ with $\operatorname{Diag} F(x) = xH'(x)$. Since

$$J(y) = \frac{y}{x(y)}\frac{dx(y)}{dy}, \qquad x(y) = \frac{y}{(1+y)^2},$$

we obtain

$$\operatorname{Diag} \widetilde{F}(y) = y\frac{d}{dy}H(x(y)).$$

The composition $H(x(y))$ is integral, so its displayed derivative has its q-th coefficient divisible by q. Equation (19) proves all gcd divisibilities. The constant and axis coefficients vanish because $A_1 \in y\mathbb{Z}[[y]]$ and $A_2 \in y^2\mathbb{Z}[[y]]$. Apply (16), and then the integral coordinate invariance, to conclude $F \in \mathcal{D}_A$. □

Corollary 9.3 (A single sufficient condition)**.** *For $n \geq 1$, put*

$$Y(n) = \sum_{l=0}^{\lfloor (n-1)/2 \rfloor} \gamma_l \binom{2i}{i}(10i+3), \qquad i = n-1-2l.$$

Then

$$\Psi \in \mathcal{D}_A \quad \Longleftrightarrow \quad n \mid Y(n) \quad (n \geq 1).$$

In particular, the latter statement alone implies the original assertion $n \mid S(n)$ for every n.

Proof. The diagonal coefficient of Ψ is exactly $Y(n)$. Indeed, the contribution with fixed l, and $i = n-1-2l$, is

$$\gamma_l \left(3\binom{2i}{i} + 20(2i-1)\binom{2i-2}{i-1}\right) = \gamma_l(10i+3)\binom{2i}{i}.$$

For $i = 0$ the second summand is absent. For $i \geq 1$, the equality uses $20(2i-1)\binom{2i-2}{i-1} = 10i\binom{2i}{i}$. Apply Theorem 9.2 with $(\alpha, \beta, R) = (3, 20, G)$, then Lemma 9.1. □

The remaining single condition can also be written

$$\int_0^x \frac{3+8t}{(1-4t)^{3/2}} G(t)\, dt \in \mathbb{Z}[[x]], \tag{20}$$

because the integrand is $\sum_{n\geq 1} Y(n)x^{n-1}$. The integral is a formal antiderivative over $\mathbb{Q}[[x]]$; its integrality is precisely the assertion to be proved, not an assumption silently used in the argument.

Corollary 9.4 (The off-diagonal condition is redundant)**.** *Assume $n \mid Y(n)$ for every $n \geq 1$. For every $1 \leq b < a$,*

$$\gcd(a,b) \;\Big|\; \sum_{l=0}^{\lfloor (b-1)/2 \rfloor} \gamma_l \binom{2(a-1-2l)}{a-1-2l}(10(a-1-2l)+3).$$

Thus condition (II) of the supplied manuscript follows from its condition (I).

Proof. By Corollary 9.3, Ψ is an integral formal divergence. Make the integral coordinate change

$$\varphi = u(1+u), \qquad \chi = \frac{v}{1+u}, \qquad E(\varphi) = (1+4\varphi)^{-1/2}.$$

The inverse of $u \mapsto u(1+u)$ is an integral Catalan series, and the logarithmic Jacobian $(1+2u)/(1+u)$ is a unit. Direct substitution shows that the same logarithmic two-form has coefficient

$$\Theta(\varphi, \chi) = \varphi\chi\, \frac{10\chi + (3-7\chi)E(\varphi)}{(1+\chi)^2} G(\varphi\chi)$$

in the new coordinates. Hence Θ is also an integral formal divergence. For $b < a$, all the indices $i = a - 1 - 2l$ occurring below are positive. Expanding $E(\varphi) = \sum_i (-1)^i \binom{2i}{i} \varphi^i$ gives exactly

$$[\varphi^a \chi^b]\Theta = (-1)^{a+b} \sum_{l=0}^{\lfloor (b-1)/2 \rfloor} \gamma_l \binom{2i}{i} (10j+3), \qquad i = a-1-2l, \quad j = b-1-2l.$$

Since $j - i = b - a$ is divisible by $\gcd(a, b)$, replacing j by i does not change this expression modulo that gcd. The coefficient itself is divisible by the gcd, by (16). This proves the claimed congruence. □

The result removes an auxiliary condition; it does not prove the remaining integrality statement (20). We have proved that this statement is equivalent to the two-variable divergence criterion and is sufficient for Sun's original conjecture. We do not assert the converse implication from the original conjecture to (20).

10 A complete unweighted branch: a quadratic invariant

The following statement is separate from the weighted all-n target. It supplies a full proof of the zero branch of Sun's Conjecture 5.8(i) [1], recorded as `OeisA179524.conjecture3` in [7]. Its historical priority is not asserted.

Theorem 10.1 (Automatic lift away from singular parameters)**.** *Let p be an odd prime and set $Y_p(c) = \sum_{r=0}^{p-1} W_r(c)$. If $c_0 \in \mathbb{Z}_p$, $c_0(c_0-1) \in \mathbb{Z}_p^\times$, and $Y_p(c_0) \equiv 0 \pmod p$, then*

$$Y_p(c_0) \equiv 0 \pmod{p^2}.$$

The binomial reductions used below are established ingredients: Sun [9, Theorem 1.1] relates this sum to a sum of Apéry polynomials, and Guo–Zeng [10, Theorem 1.3] give the binomial reduction used here. We include its direct proof to make the subsequent differential argument checkable without additional files. The finite-invariant lifting strategy itself has a direct antecedent in Babanskyy [11, Lemmas 2.1–2.2]: a quadratic differential invariant, explicit treatment of exceptional degree-p coefficients, and a double-root argument yield a lift from modulo p to modulo p^2. Here that strategy is adapted to the dual polynomial below. The proof is given directly; the invariant principle is not claimed as a new invention.

Lemma 10.2. *For an even integer m with $0 \le m \le p-1$,*

$$\sum_{r=m}^{p-1} \binom{r}{m}^2 \equiv \binom{p+m}{2m+1} \pmod{p^2}.$$

Proof. For $m \le r < p$, all denominators in $H_r - H_{r-m}$ are p-adic units, and

$$\binom{p+m-1-r}{m} \equiv \binom{r}{m}[1 - p(H_r - H_{r-m})] \pmod{p^2}.$$

The sign is positive because m is even. The involution $r \mapsto p+m-1-r$ permutes $m, \dots, p-1$, preserves $\binom{r}{m}^2$ modulo p, and reverses $H_r - H_{r-m}$ modulo p. The last fact follows from $H_{p-1-u} \equiv H_u \pmod p$. Thus the sum of the first-order corrections is zero modulo p. Vandermonde's identity gives

$$\sum_{r=m}^{p-1} \binom{r}{m} \binom{p+m-1-r}{m} = \binom{p+m}{2m+1},$$

proving the claim, including $m = 0$ and $m = p-1$. □

Lemma 10.3 (Dual polynomial and differential equation)**.** *Put $h = (p-1)/2$ and*

$$Z_p(c) = \sum_{s=0}^{h} v_s c^s, \qquad v_s = p\frac{(2s)!^4}{s!^4(4s+1)!}.$$

Then $v_s \in \mathbb{Z}_p$, $Y_p \equiv Z_p \pmod{p^2}$ coefficientwise, and, with $\theta = c\,d/dc$,

$$\mathcal{L}Y_p \equiv 0 \pmod{p^2}, \qquad \mathcal{L} = \theta(4\theta-1)(4\theta+1) - 2c(2\theta+1)^3. \tag{21}$$

Proof. The denominator $(4s+1)!$ contains at most one factor p, since $4s+1 \le 2p-1$; it is canceled by the displayed factor p. Lemma 10.2 makes the coefficient of c^s in Y_p congruent to

$$\binom{2s}{s}^2\binom{p+2s}{4s+1} = p\binom{2s}{s}^2\frac{\prod_{j=1}^{2s}(p^2-j^2)}{(4s+1)!}.$$

The product is $(2s)!^2$ modulo p^2, since $2s$ is even. Before division the error has valuation at least three; the factorial denominator loses at most one. Hence the error is still zero modulo p^2, proving the coefficientwise congruence.

The exact coefficient relation is

$$s(4s-1)(4s+1)v_s = 2(2s-1)^3 v_{s-1} \qquad (1 \le s \le h).$$

Consequently the entire truncation residual is

$$\mathcal{L}Z_p = -2p^3 v_h c^{h+1}.$$

It vanishes modulo p^3. Applying the integer differential operator to $Y_p - Z_p$ proves (21). No coefficient s or $4s \pm 1$ has been inverted modulo p. □

Lemma 10.4 (Quadratic differential invariant)**.** *For a polynomial Y, define*

$$\mathcal{J}[Y] = 16c^2(c-1)(Y'^2 - 2YY'') - 16c(3c-2)YY' - (4c+1)Y^2.$$

Then

$$\frac{d}{dc}\mathcal{J}[Y] = 2Y\,\frac{\mathcal{L}Y}{c}.$$

In particular, $\mathcal{J}[Y_p] \equiv 0 \pmod{p^2}$ coefficientwise.

Proof. The differential identity follows by expanding

$$\frac{\mathcal{L}Y}{c} = 16c^2(1-c)Y''' + (48c - 72c^2)Y'' + (15 - 52c)Y' - 2Y.$$

The right side is polynomial. Thus $\mathcal{J}[Y_p]'$ is zero modulo p^2. A zero derivative alone would not suffice, because a degree-p term could survive. We check that term explicitly.

Since $\deg Y_p = h$, the degree of $\mathcal{J}[Y_p]$ is at most $2h+1 = p$. Its constant coefficient is $-Y_p(0)^2 = -p^2$. If K is the leading coefficient of Y_p, its degree-p coefficient is

$$[16(h^2 - 2h(h-1)) - 48h - 4]K^2 = -4(2h+1)^2K^2 = -4p^2K^2.$$

Both endpoints vanish modulo p^2. For every intermediate index $1 \le j \le p-1$, the derivative gives $j\,[c^j]\mathcal{J}[Y_p] = 0$ modulo p^2, and j is a unit. All coefficients therefore vanish. □

Proof of Theorem 10.1. Equation (11) gives

$$Y_p(c) \equiv c^h P_d(t)^2 \pmod{p}, \qquad d = \lfloor p/4 \rfloor, \quad t^2 = 1 - 1/c.$$

Choose t_0 over the algebraic closure of the residue field with $t_0^2 = 1 - 1/c_0$. The assumptions make $t_0 \neq 0$, $t_0^2 \neq 1$, and $P_d(t_0) = 0$. Such a Legendre root is simple. Indeed, if its multiplicity were $m \geq 2$, the lowest-degree term of

$$(1 - t^2)P_d'' - 2tP_d' + d(d+1)P_d = 0$$

would have a nonzero coefficient proportional to $(1 - t_0^2)m(m-1)$, since $m \leq d < p$. The degree-zero case has no roots and is vacuous.

Because $dt/dc = 1/(2tc^2)$, differentiating the polynomial congruence gives

$$Y_p'(c_0) \equiv 0, \qquad Y_p''(c_0) \equiv 2c_0^h P_d'(t_0)^2 (2t_0 c_0^2)^{-2} \neq 0 \pmod{p}.$$

The expression is in the base residue field, as is the left side; the field extension is used only to describe the root.

Evaluate the invariant at c_0. Both $Y_p(c_0)$ and $Y_p'(c_0)$ are divisible by p, so every term except the Y_pY_p'' term vanishes modulo p^2. We obtain

$$32c_0^2(c_0 - 1)Y_p(c_0)Y_p''(c_0) \equiv 0 \pmod{p^2}.$$

Its coefficient multiplying $Y_p(c_0)$ is a p-adic unit. This forces $Y_p(c_0) \equiv 0 \pmod{p^2}$. □

Corollary 10.5 (Sun's inert unweighted branch)**.** *For every prime* $p \equiv 11, 13, 17, 19 \pmod{20}$,

$$\sum_{r=0}^{p-1} a_r \equiv 0 \pmod{p^2}.$$

Proof. The CM argument in Theorem 5.1 gives $Y_p(-4) = 0 \pmod{p}$. Here $c_0 = -4$ satisfies $c_0(c_0 - 1) = 20$, a unit at all the specified primes. Apply Theorem 10.1. □

The corollary closes all quantifiers of this particular unweighted branch. It neither proves the split-prime evaluation branches of Sun's Conjecture 5.8(i), nor the weighted all-n statement (2) in Conjecture 5.8(ii).

11 A direct symbolic proof modulo 9

Proposition 11.1. *For every* $m \geq 1$, $9 \mid S(9m)$.

Proof. The digit expansion (7) at $p = 3$, or a direct factorial expansion, gives for every $q \geq 0$

$$a_{3q} \equiv a_q + 6L_q, \qquad a_{3q+1} \equiv a_q, \qquad a_{3q+2} \equiv -15a_q + 3L_q \pmod{9}. \tag{22}$$

Here is an explicit check of the coefficients. The only low-digit pairs with $2s \leq r$ and their corresponding weights are

$$(r, s) = (0,0), (1,0), (2,0), (2,1), \qquad f(r,s) = 1, 1, 1, -16.$$

Use $H_0 = 0, H_1 = 1, H_2 = 3/2$, together with

$$(-4)^{3j+s} \equiv (-4)^{j+s}(1 + 6j) \pmod{9}.$$

The correction divided by 3 is

$$2q(H_r - H_{r-2s}) + 4j(H_{r-2s} - H_s) + 2j.$$

Its q-coefficient is zero modulo 3 in all four cases, and its j-coefficients are $2, 0, 2, 1$, respectively. All low-carry terms vanish modulo 9; higher carries remain in $M(q, j)$. This proves (22) for arbitrary q.

Consequently

$$\sum_{r=0}^{2}(20(3q+r)+17)a_{3q+r} \equiv 3(qa_q + L_q) \pmod 9.$$

Modulo 3, Lucas gives $a_{3u+r} \equiv a_u a_r$ and $L_{3u+r} \equiv a_u L_r$. Since

$$(a_0, a_1, a_2) \equiv (1, 1, 0), \qquad (L_0, L_1, L_2) \equiv (0, 0, 2) \pmod 3,$$

we have

$$\sum_{r=0}^{2}((3u+r)a_{3u+r} + L_{3u+r}) \equiv a_u \sum_{r=0}^{2}(ra_r + L_r) = 0 \pmod 3.$$

Sum the preceding modulo-9 block over $q < 3m$, and group the right side into these three-term blocks. □

12 Certified computation and the remaining bottom layers

Appendix A proves an exact telescoping certificate for the order-four recurrence stated in the supplied manuscript. The leading recurrence coefficient never vanishes for $n \geq 0$, so the initial values determine the original binomial-sum sequence uniquely. Appendix B then computes this sequence with exact integer arithmetic, checking each division. The fresh run verifies $n \mid S(n)$ for every $1 \leq n \leq 50000$, with no failure.

Every odd prime power $p^e \leq 50000$ comes with its complete chain of smaller bases. Theorem 8.2 therefore turns each such base into $p^e \mid S(p^e m)$ for all $m \geq 1$. Together with Corollaries 6.5, 7.2, Theorem 3.1, and the prime case, this proves Theorem 1.1.

Some fresh exact checks of the analytic conclusions are recorded below. They are checks on the implementation and are not assumptions of the unbounded-exponent proofs. Put $M_2(N) = \sum_{n<N} n^2 a_n$.

p	e with $N = p^e$	$v_p(U(N))$	$v_p(T(N))$	$v_p(S(N))$	$v_p(M_2(N))$
13	3	4	3	3	2
17	3	4	4	4	2
19	3	4	3	3	2
11	4	4	5	4	4
5	6	6	6	6	6
13	4	4	5	4	4

For comparison with earlier formulations, define the Newton coefficients

$$b_\ell = \sum_{j=0}^{\ell}(-1)^{\ell-j}\binom{\ell}{j}a_j, \qquad b_{-1} = 0.$$

Binomial inversion and (4) give

$$\frac{S(n)}{n} = \sum_{\ell<n}\binom{n-1}{\ell}\frac{(20\ell+17)b_\ell + 20\ell b_{\ell-1}}{\ell+1}.$$

Since the transform and its inverse are integral, the original conjecture is equivalent to

$$\ell + 1 \mid 3b_\ell + 20b_{\ell-1} \qquad (\ell \geq 0).$$

This remains an equivalent unsolved condition. The single sufficient integrality statement in (20) is also not proved.

The remaining original cases are bottom values $p^e \mid S(p^e)$ at split primes $p \equiv 1, 3, 7, 9 \pmod{20}$, beyond their displayed finite exponent bounds. There is now no missing all-cofactor propagation step once a finite chain of bottom values is supplied. Neither the two nonzero unweighted evaluations in Sun 5.8(i) nor the separate weighted prime-square refinement

$$S(p) \stackrel{?}{\equiv} p\left(7 + 10\left(\frac{-1}{p}\right)\right) \pmod{p^2}$$

is proved in general here.

13 Provenance, review, and scope

The exact original target is the statement recorded as `OeisA179524.conjecture4` in the inspected formal-conjectures file [7]. Its first path commit is dated 20 August 2026, rather than the November date given in the supplied manuscript. Although the OEIS Open article [8] does not mention this sequence by its identifier, the released benchmark includes the target. These metadata observations are not a certificate that no later solution exists.

The supplied manuscript [12] establishes useful exact reductions but states its order-four recurrence as empirically found. Appendix A supplies the missing all-index proof. Its reported million-index run and prime cutoff 6000 are not inherited here: the cited external computation files were not among the attachments. The independently reproduced cutoff in this note is 50000, and its complete program is embedded in the single source.

The use of analytic factorial weights and a simultaneous induction with a one-power loss for arbitrary test functions has a substantial antecedent in Delaygue [13, Theorem 1, Proposition 1, and Sections 3.3–4.3]. Here we give an explicit restricted-series operator for the triangular truncations in Sun's problem, an affine first-order refinement, full carry control, and moment criteria implying the stated prime-power divisibilities. No claim is made to have invented the general analytic-weight strategy. The finite differential-invariant method used for the separate unweighted branch also has the explicit antecedent cited in Section 10.

Independent mathematical derivations checked the exact interpolation, the Gauss-norm bounds, every carry boundary, the simultaneous induction, the $p = 5$ coefficient table, all-layer propagation, the Catalan change of variables, and the recurrence certificate. This is internal mathematical review, not external peer review or a Lean formalization. A bounded primary-source search did not identify an exact earlier resolution of the new target-specific conclusions; it does not establish historical nonduplication. Thus the full original all-n success condition remains unmet, and historical priority remains unasserted.

A An exact certificate for the order-four recurrence

This appendix supplies a complete telescoping certificate for the order-four recurrence stated in the supplied manuscript. In particular, the recurrence is established for the original binomial sum at every nonnegative index; it is not assumed on the basis of numerical agreement. Write

$$f(n,k) = \begin{cases} (-4)^k \dfrac{(n!)^2}{(k!)^4((n-2k)!)^2}, & 0 \le 2k \le n, \\ 0, & 2k > n, \end{cases} \qquad a(n) = \sum_{k \ge 0} f(n,k). \tag{23}$$

All indices in this appendix are nonnegative integers unless stated otherwise.

A.1 The recurrence and its certificate

Define the following five polynomials:

$$p_0(n) = -4225(n+1)^2(n+2)^2(240n^3+2160n^2+6483n+6490), \tag{24}$$

$$p_1(n) = -2(n+2)^2\big(91680n^5+1146000n^4+5647266n^3 + 13701286n^2+16361708n+7698385\big), \tag{25}$$

$$p_2(n) = -2\big(16080n^7+289440n^6+2219001n^5+9390796n^4 + 23689644n^3+35620787n^2+29562269n+10448048\big), \tag{26}$$

$$p_3(n) = 2(n+3)\big(480n^6+7920n^5+53646n^4+190760n^3 + 375516n^2+388401n+165272\big), \tag{27}$$

$$p_4(n) = -(n+3)(n+4)^3(240n^3+1440n^2+2883n+1927). \tag{28}$$

For a compact certificate, put

$$A_n = 240n^3+2160n^2+6483n+6490,$$

$$B_n = 240n^3+1440n^2+2883n+1927,$$

$$C_n = 720n^5+8640n^4+48369n^3+163272n^2+315171n+257123, \tag{29}$$

$$P_n(t) = 65A_nt^4+4(n-31)A_nt^3+2C_nt^2+(n+3)^3B_n(4t+n+4), \tag{30}$$

$$G(n,k) = \frac{16k^4}{(n+3)^2(n+4)^2}P_n(n+4-2k)f(n+4,k). \tag{31}$$

The summand identity is

$$\sum_{i=0}^{4} p_i(n)f(n+i,k) = G(n,k+1)-G(n,k). \tag{32}$$

Consequently, the original sequence satisfies

$$\boxed{\sum_{i=0}^{4} p_i(n)a(n+i) = 0 \qquad (n\geq 0).} \tag{33}$$

A.2 Polynomial verification and the endpoints

Here is a verification of (32) that includes the truncated summation boundaries. Set

$$D_n = \prod_{j=1}^{4}(n+j)^2,$$

$$Q(n,k) = \sum_{i=0}^{4} p_i(n)\prod_{j=1}^{i}(n+j)^2\prod_{j=i+1}^{4}(n+j-2k)^2, \tag{34}$$

$$R(n,k) = 16(n+1)^2(n+2)^2P_n(n+4-2k). \tag{35}$$

Empty products are interpreted as 1. Substitution of (24)–(28) and (29)–(30) gives the polynomial identity in $\mathbb{Z}[n,k]$

$$\boxed{\begin{aligned}Q(n,k) = &-4(n+4-2k)^2(n+3-2k)^2R(n,k+1)\\ &-k^4R(n,k).\end{aligned}} \tag{36}$$

Thus the entire certificate can be checked by finite polynomial multiplication and coefficient comparison. This is an identity of polynomials, rather than a test at finitely many numerical pairs.

For $0 \le 2k \le n+4$ one has

$$D_n f(n+i,k) = f(n+4,k) \prod_{j=1}^{i} (n+j)^2 \prod_{j=i+1}^{4} (n+j-2k)^2. \tag{37}$$

When $2k \le n+i$, this follows directly by cancelling factorials. When $n+i < 2k \le n+4$, the left-hand side is zero by definition; the right-hand side is also zero because its second product contains the factor with $j = 2k-n$. This proves (37) even when some of the five shifted summands have left their support.

The remaining ratio is

$$f(n+4,k+1) = -4\frac{(n+4-2k)^2(n+3-2k)^2}{(k+1)^4} f(n+4,k). \tag{38}$$

It holds throughout the support of $f(n+4,k)$, including its last nonzero term: at that term either $n+4-2k=0$ or $n+3-2k=0$, and both sides of (38) are zero. Multiplying (36) by $f(n+4,k)/D_n$, and using (37) and (38), proves (32) for $0 \le 2k \le n+4$. For $2k > n+4$ all its terms vanish. Finally, $G(n,0) = 0$ because of the factor k^4, and $G(n,k) = 0$ for sufficiently large k by finite support. Summing (32) over $k \ge 0$ therefore proves (33) with no boundary remainder.

A.3 What the certificate does and does not establish

For every $n \ge 0$ the leading coefficient $p_4(n)$ is nonzero. Together with

$$(a(0), a(1), a(2), a(3)) = (1, 1, -15, -143), \tag{39}$$

the recurrence therefore determines the original binomial-sum sequence uniquely. A subsequent calculation using this recurrence and exact integer arithmetic consequently computes that original sequence, provided the implementation and the stated range are independently checked.

The certificate alone does not authenticate the supplied manuscript's claim of a check through one million indices. The computation itself has not been recovered from that claim, and the external files `ctest.c` and `verify_log.txt` mentioned there were not included in the supplied TEX and PDF. Any finite-range verification reported in the present paper must therefore be supported by a fresh computation or by an independently checked log. Neither a recurrence certificate nor a finite computation proves $N \mid S(N)$ for all N.

B Complete exact verification program

The following Python 3.11-or-later program uses only the standard library. The recurrence is justified by Appendix A; comparison with the original definition through index 100 checks its implementation. All 49996 recurrence divisions and 50000 divisibility tests passed in the fresh run. The additional checkpoint valuations and exact density comparison are reproduced by the same program. The analytic all-exponent theorems do not depend on this computation.

```
from pathlib import Path
from math import comb, isqrt
import sys, time, json
if hasattr(sys, "set_int_max_str_digits"):
    sys.set_int_max_str_digits(0)
N = 50000
def ps(n):
    return (
```

```
    -4225*(n+1)**2*(n+2)**2*(240*n**3+2160*n**2+6483*n+6490),
    -2*(n+2)**2*(91680*n**5+1146000*n**4+5647266*n**3+13701286*n**2+16361708*n+7698385),
    -2*(16080*n**7+289440*n**6+2219001*n**5+9390796*n**4+23689644*n**3+35620787*n**2+29562269*n
    +10448048),
    2*(n+3)*(480*n**6+7920*n**5+53646*n**4+190760*n**3+375516*n**2+388401*n+165272),
    -(n+3)*(n+4)**3*(240*n**3+1440*n**2+2883*n+1927))
def direct(n):
    return sum((-4)**k*comb(n,k)**2*comb(n-k,k)**2 for k in range(n//2+1))
def val(x,p):
    if not x:return None
    v=0
    while x%p==0:x//=p;v+=1
    return v
win=[1,1,-15,-143]
S=U=T=M2=0
checkpoints={15625:(5,6),14641:(11,4),28561:(13,4),2197:(13,3),4913:(17,3),6859:(19,3)}
rows=[]
start=time.monotonic()
for n in range(N):
    if n<4: an=win[n]
    else:
        P=ps(n-4)
        an,rem=divmod(-sum(P[i]*win[i] for i in range(4)),P[4])
        assert rem==0
        win=win[1:]+[an]
    if n<=100: assert an==direct(n)
    U+=an;T+=n*an;M2+=n*n*an;S+=(20*n+17)*an
    assert S%(n+1)==0,("counterexample",n+1)
    if n+1 in checkpoints:
        p,e=checkpoints[n+1]
        row={"N":n+1,"p":p,"e":e,"vU":val(U,p),"vT":val(T,p),"vS":val(S,p),"vM2":val(M2,p)}
        assert row["vU"]>=e and row["vS"]>=e
        if p!=5: assert row["vT"]>=e and row["vM2"]>=e-1
        rows.append(row)
    if (n+1)%10000==0:print("checked",n+1,"elapsed",round(time.monotonic()-start,2),flush=True)
result={"verified_through":N,"all_divisibilities_passed":True,"recurrence_divisions_exact":True,
    "direct_comparison_through":100,"checkpoints":rows,"seconds":round(time.monotonic()-start,3)
    }
print(json.dumps(result,indent=2),flush=True)
Path(__file__).with_suffix(".json").write_text(json.dumps(result,indent=2)+"\n")


from fractions import Fraction
R = Fraction(1)
for p in range(3,224,2):
    if not all(p%d for d in range(2,isqrt(p)+1)): continue
    e=0; power=1
    while power*p<=N: power*=p; e+=1
    good=(p==5 or p%20 in (11,13,17,19))
    nu=Fraction(1) if good else 1-Fraction(1,p**(e+1))
    R*=nu/(1-Fraction(1,p*p))
assert 8*R/Fraction(355,113)**2 > Fraction(9992,10000)
print("Exact density lower bound > 0.9992: PASS")
```

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